

Cascode Source Compensation in a Differential Class AB DDA

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Abstract—The supplied differential difference amplifier (DDA) has differential outputs, a class AB output stage, and four series RC compensation branches. Moving each capacitor's first-stage terminal from a cascode drain to the corresponding source can remove the need for a resistor whose purpose is to suppress the ordinary Miller right-half-plane zero. The amplifier-output terminal of each capacitor stays connected to its original output. This is cascode, or indirect, compensation. A simplified small-signal model explains the zero's removal and the additional dynamics introduced by the source node. The proposed node connections are inferred from a screenshot; no netlist, operating-point data, or simulations were available. Stability of the specific DDA remains unverified.

Index Terms—Differential difference amplifier, class AB, cascode compensation, indirect compensation, Miller compensation.

I. CONCLUSION AND SCOPE

YES, the series resistors can potentially be omitted by returning the compensation current to the sources of the *first-stage* cascodes. The cascodes then act as common-gate current buffers. This implements the principle of removing the direct feedforward path while retaining output-dependent feedback current [1], [2].

The conclusion concerns the resistors' zero-nulling function. It does not establish that the existing capacitor values will give sufficient phase margin after relocation. Resistors selected to provide additional phase lead or damping may still have a useful role. The proposed change is therefore a topology to evaluate, rather than a validated replacement.

The evidence is the user-supplied screenshot dated October 9, 2026, showing Fig. 9 of `IN_AMP_GF180.pdf`. The discussion assumes that the four visible RC branches connect each output to its PMOS and NMOS gate-driving nodes. Component values and some labels are insufficiently clear to support a quantitative design calculation.

II. WHY DIRECT MILLER COMPENSATION USES A RESISTOR

Let v_1 drive a simple inverting common-source output stage, and let v_o be its output. A capacitor C_c directly between these nodes supplies a feedforward current proportional to $sC_c v_1$, which opposes the output-stage current $g_{m2}v_1$. Their cancellation creates a right-half-plane (RHP) zero. With series resistance R_z , the corresponding simplified result is

$$s_{z,\text{direct}} = \frac{g_{m2}}{C_c(1 - g_{m2}R_z)}. \quad (1)$$

Thus $R_z = 0$ gives $s_z = +g_{m2}/C_c$; $R_z = 1/g_{m2}$ sends this idealized zero to infinity; and larger R_z places it in the left half-plane (LHP) [1].

Equation (1) describes one effective output path. It must not be applied independently to the four branches of this DDA without accounting for the coupling between the two class AB gate-driving nodes.

III. HOW THE CASCODE CHANGES THE COMPENSATION PATH

Let v_x denote a first-stage cascode source and v_1 its drain, which drives the output stage. Relocate C_c from (v_o, v_1) to (v_o, v_x) . For a saturated cascode with an AC-stiff gate and bulk, its source transconductance is approximately

$$G_c = g_{m,\text{cas}} + g_{mb,\text{cas}}. \quad (2)$$

In the unilateral approximation, the source-node impedance is of order $1/G_c$. Output motion injects capacitor current at this source; the cascode conveys it to the high-impedance drain. Drain voltage motion no longer directly drives the compensation capacitor. The internal common-gate device provides the current-buffer function [2].

A. An Explicit Illustrative Model

The following three-node model makes the distinction precise. Let G_1, C_1 be the conductance and capacitance at the gate-driving node, G_o, C_o those at the output, and C_x the source-node parasitic capacitance. C_o includes the load but excludes C_c . Inject first-stage signal current i_1 at v_1 . Neglect cascode drain-source conductance, intrinsic feedthrough capacitances, and source-node shunt conductance. Choose the current-buffer polarity that gives negative compensation feedback. Nodal equations are

$$(G_1 + sC_1)v_1 - G_c v_x = i_1, \quad (3)$$

$$[G_c + s(C_x + C_c)]v_x - sC_c v_o = 0, \quad (4)$$

$$g_{m2}v_1 + [G_o + s(C_o + C_c)]v_o - sC_c v_x = 0. \quad (5)$$

Define

$$A(s) = G_c + s(C_x + C_c), \quad (6)$$

$$Y_1(s) = G_1 + sC_1, \quad (7)$$

$$Y_o(s) = G_o + s(C_o + C_c). \quad (8)$$

Solving gives

$$\frac{v_o}{i_1} = \frac{-g_{m2}A(s)}{Y_1(s)[Y_o(s)A(s) - s^2C_c^2] + g_{m2}G_c sC_c}. \quad (9)$$

TABLE I
PROVISIONAL FIRST STAGE TERMINAL CHANGES FOR V_{o^-}

Branch	Present node	Proposed node
Upper	n10	n7, source of the associated PMOS cascode biased by vb2
Lower	Vofcp	Source of the associated NMOS cascode biased by vb4, directly below the present node

The numerator's zero is

$$s_{z,\text{indirect}} = -\frac{G_c}{C_x + C_c}, \quad (10)$$

which is in the LHP for positive capacitances and G_c . The denominator is generally cubic. Consequently, removing the direct Miller RHP zero introduces source-node dynamics rather than guaranteeing a two-pole response. This agrees with the standard indirect-compensation description [2].

Equations (9) and (10) are an illustrative reduction, not a transfer function extracted from the DDA. In the actual circuit, signal current may also enter a cascode source, changing the numerator. Finite cascode output conductance, bias impedance, and transistor capacitances restore residual reverse transmission. Therefore neither the exact zero locations nor the source impedance of the complete circuit can be inferred from this model alone.

IV. PROPOSED CONNECTIONS IN THE SUPPLIED SCHEMATIC

For the V_{o^-} half, the screenshot suggests the changes in Table I. Both output-side capacitor terminals remain at V_{o^-} . Remove the series resistors for the initial indirect-compensation trial and apply the corresponding changes to the V_{o^+} half.

These names are a provisional visual reading. Confirm connectivity and transistor source/drain assignments in the original schematic or netlist before editing. Each return must pass through the cascode associated with the original gate-driving node and preserve the local feedback polarity.

V. VERIFICATION REQUIRED FOR THIS DDA

At quiescence, the PMOS and NMOS output devices both contribute transconductance. Their gate nodes are coupled through the class AB bias circuit, so a single g_{m2} does not capture all internal modes. Under substantial source or sink

load, one output device can dominate. The relevant operating points should therefore include zero load and both signs of DC load current across the intended output swing.

Evaluate differential return ratio with the common-mode feedback (CMFB) loop active. Evaluate CMFB return ratio with the intended differential feedback connected. Inspect all unity-gain crossings and internal modes; a favorable margin at one crossing is insufficient if additional crossings or poorly damped modes exist. Repeat across the specified load capacitance, supply, process, temperature, and common-mode ranges.

Use small steps to check settling and ringing, and large steps in both directions to check slew rate and recovery. In indirect compensation, the cascode branch must carry compensation current of approximate magnitude

$$|i_c| \simeq C_c \left| \frac{dv_o}{dt} \right| \quad (11)$$

while its source remains nearly stationary. If available branch current or headroom is exhausted, the small-signal buffer approximation fails. Class AB output drive alone does not eliminate this limitation [3].

VI. DESIGN DECISION

Cascode-source feedback is a technically justified way to remove the ordinary Miller zero-nulling resistors. Confirm the proposed nodes, remove the resistors in a simulation variant, and retune the capacitors against the full circuit's differential and common-mode dynamics. The screenshot supports the topology change; it does not establish acceptable stability or settling for the modified amplifier.

REFERENCES

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